

Tax Evasion and Productivity

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Recap

Last meeting: What is an interesting question I can answer with what I have?

- ▶ Geographic variation in tax evasion
- ▶ Counterfactual question: What would have been the optimal tax rate for the government?

Today:

- ▶ Geographic variation: what are the variables in the data, what do they look like?
- ▶ Optimal tax rate: strategy for the simplest case

Introduction

- ▶ What's the question?
- ▶ Why should you care?

Contribution

- ▶ What do I do?
- ▶ How do I do it?

Overview of Results

- ▶ What do I find?
- ▶ Why is it important?

Model

$$\begin{aligned} \max_{K_{it}, L_{it}, M_{it}, e_{it}} \quad & (1 - q(e_{it}, a)) \left[P_t \mathbb{E}_\varepsilon[Y_{it}] - \rho_t M_{it} - r_t K_{it} - w_t L_{it} \right. \\ & \left. - \tau (P_t \mathbb{E}_\varepsilon[Y_{it}] - \rho_t (M_{it} + e_{it})) \right] \\ & + q(e_{it}, a) \left[(1 - \tau) (P_t \mathbb{E}_\varepsilon[Y_{it}] - \rho_t M_{it}) - r_t K_{it} - w_t L_{it} \right. \\ & \left. - \phi \rho_t \tau e_{it} \right] - \kappa_i(e_{it}) \\ \text{s.t.} \quad & Y_{it} = K_{it}^{\alpha_k} L_{it}^{\alpha_L} M_{it}^\beta \exp(\omega_{it} + \varepsilon_{it}) \end{aligned} \tag{1}$$

Important Assumptions

- ▶ Following seminal work of (**Allingham1972?**), tax authority audits firms at random and always detects tax evasion
 - ▶ The probability of detection, $q(e_{it}, a)$, is a function of the level of tax evasion, e_{it} , and the monitoring technology, a , of the tax authority.
 - ▶ Need to review in light of empirical evidence. Empirical model assumes that choosing M does not enter the probability of detection or the cost of evasion. In other words, choosing M is independent of choosing e .
- ▶ The cost of evasion: any resources of the firm allocated to the tax evasion activity that could have been employed in a productive activity (e.g. manager's time spent on tax evasion instead of managing the firm).
 - ▶ Unlike the penalty of tax evasion, ϕ , it does not depend on whether you are detected or not
 - ▶ Increasing in the level of tax evasion, e_{it}

Model (Cont.)

The firms' problem (Equation 1) can be rewritten as follows,

$$\begin{aligned} \max_{K_{it}, L_{it}, M_{it}, e_{it}} \quad & \mathbb{E}[\pi_{it} | \Gamma] = (1 - \tau) (P_t \mathbb{E}[Y_{it}] - \rho_t M_{it}) - r_t K_{it} - w_t L_{it} \\ & + [1 - q(e_{it}, a)] (\rho_t \tau e_{it}) - q(e_{it}, a) \phi \rho_t \tau e_{it} - \kappa_i(e_{it}) \\ \text{s.t.} \quad & Y_{it} = K_{it}^{\alpha_K} L_{it}^{\alpha_L} M_{it}^{\beta} \exp(\omega_{it} + \varepsilon_{it}) \end{aligned} \tag{2}$$

First Order Condition (FOC)

An interior solution leads to following first-order conditions,

$$\beta K_{it}^{\alpha_K} L_{it}^{\alpha_L} M_{it}^{\beta-1} \exp(\omega_{it}) \mathcal{E} = \frac{\rho_t}{P_t} \quad (3)$$

$$\tau \rho_t \left(1 - (1 + \phi) \left(q(e_{it}, a) + \frac{\partial q(e_{it}, a)}{\partial e_{it}} e_{it} \right) \right) = \kappa'_i(e_{it}) \quad (4)$$

where $\mathcal{E} = \mathbb{E}[\exp(\varepsilon_{it})]$.

Tax Evasion

To better understand the incentives to overreport inputs, we can rewrite Equation 4 in terms of the evasion elasticity of the probability of detection,

$\xi_{q,e} = \frac{\partial q(e_{it}, a)}{\partial e_{it}} \frac{e_{it}}{q(e_{it}, a)}$, as follows,

$$\tau \rho_t \left(1 - (1 + \phi) (1 + \xi_{q,e}) q(e_{it}, a) \right) = \kappa'_i(e_{it}) \quad (5)$$

Corner Solution

Note that there could be firms that can optimally choose not to evade, $e_{it} = 0$.

For these firms, the marginal cost of tax evasion is greater than the marginal benefit for any non-zero level of tax evasion.

This implies that for these firms, Equation 4 becomes,

$$\tau\rho_t \leq \kappa'_i(0) \tag{6}$$

Empirical Strategies

- ▶ Testing for the presence of tax evasion by industry
 - ▶ Robust to different specifications
- ▶ Regression
 - ▶ Useful to study tax evasion heterogeneity across firms and over time
- ▶ Deconvolution
 - ▶ Recover distribution of tax evasion and productivity
 - ▶ Estimation of unbiased Production Function

Data

Colombia

Testing

Tax Evasion Over Time

Incentives to evade taxes by cost overreporting changed over time due to changes in fiscal policy

The reform of 1983 increased, from 6% to 10%, the Sales Tax (ST) rate of most industries.

Empirical Approach

Taking the *FOC* and stacking over industries. For Corporations, the truth-reporters, i in industry j at year t , we have,

$$s_{ijt}^C = \gamma_j - \varepsilon_{ijt} \quad (7)$$

Where s_{ijt}^C is the share of materials' cost over sales, γ_j is the output elasticity of materials for industry j (CD-PF), and ε_{ijt} is the error term

Unincorporated Firms

Unincorporated firms, the evaders, we have,

$$s_{ijt}^N = \gamma_j + e_{ijt} - \varepsilon_{ijt} \quad (8)$$

Where e_{ijt} is the overreporting of materials' cost due to tax evasion.

Combining Corporations and Unincorporated Firms

Let D_{ijt}^N be a dummy indicating whether the firm is an unincorporated firm

$$\begin{aligned}(1 - D_{ijt}^N)s_{ijt} + D_{ijt}^Ns_{ijt} &= (1 - D_{ijt}^N)(\gamma_j - \varepsilon_{ijt}) \\ &\quad + D_{ijt}^N\left(\gamma_j + e_{ijt} - \varepsilon_{ijt}\right) \\ s_{ijt} &= \gamma_j + D_{ijt}^Ne_{ijt} - \varepsilon_{ijt}\end{aligned}\tag{9}$$

Taking Expectations

$$E[s_{ijt}|D^N = 0] = \gamma_j \quad (10)$$

$$E[s_{ijt}|D^N = 1] - E[s_{ijt}|D^N = 0] = \mu$$

we can identify the average overreporting level, μ

Changes Over Time

In the case of the fiscal policy changes, we might be interested in the changes in average tax evasion after the 1983 fiscal reform.

We can take the difference of the expectations between year t' and $t = 1983$.

$$\begin{aligned} s_{ijt'} - s_{ijt=1983} &= D_{ijt}^N (e_{ijt'} - e_{ijt=1983}) - (\varepsilon_{ijt'} - \varepsilon_{ijt=1983}) \\ E[s_{ijt'} - s_{ijt=1983} | D^N = 1] &= \mu_{t'} - \mu_{t=1983} \\ &\equiv \Delta_{t'} \end{aligned} \tag{11}$$

μ_t is the annual average tax evasion of year t , and Δ_t is the change in average tax evasion between year t and 1983, the fiscal reform year.

In Regression Form

To estimate Equation 9, I use the following regression specification,

$$s_{ijt} = \gamma_j + \sum_{t=1981}^{1991} \mu_t D_{ijt}^N + \eta_{ijt} \quad (12)$$

- ▶ Where μ_t are year dummies,
 - ▶ μ_t is the average tax evasion of unincorporated firms in year t
- ▶ γ_j , the industry fixed effects, captures the output elasticity of materials of each industry
- ▶ D_{ijt}^N and are defined as before.
- ▶ $\eta_{ijt} = \xi_{ijt} - \varepsilon_{ijt}$, where $e_{ijt} = \sum_t \mu_t + \xi_{ijt}$, and ξ_{ijt} is the idiosyncratic component of tax evasion that is not captured by the year dummies

In Regression Form: Differences Over Time

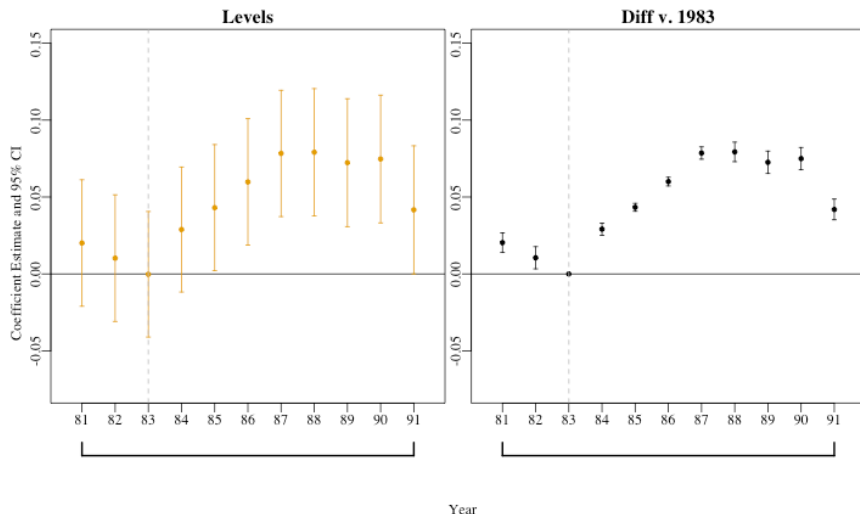
Likewise, the following regression specification is employed to estimate Equation 11,

$$s_{ijt} = \mu D_{ijt}^N + \sum_{t=1981}^{1991} \mathbb{1}[t \neq 1983] \Delta_t D_{ijt}^N + \gamma_j + \eta_{ijt} \quad (13)$$

- ▶ Where Δ_t are year dummies,
 - ▶ Δ_t is the difference in tax evasion of unincorporated firms between year t and 1983, the fiscal reform year
- ▶ μ , the average tax evasion of unincorporated firms in 1983
- ▶ γ_j , D_{ijt}^N and η_{ijt} are defined as before.

Tax Evasion Increases after the Fiscal Reform of 1983

Tax Evasion by Unincorporated Firms Overall Input Overreporting Across All Industries



Results

- ▶ After the fiscal reform of 1983 increased the sales tax rate, tax evasion through input overreporting increased gradually
- ▶ Input overreporting reached an average of 8% in 1987, where it stabilized until 1990

Why Do The Standard Errors Look So Different?

- ▶ **Levels:** The large error terms suggest that tax evasion is widely heterogeneous across firms
 - ▶ Where does this heterogeneity come from? Differences in the tax rate due to firm characteristics (industry, JO, and exports), the (perceived) probability of detection (model assumes risk neutrality, but firms might have different perceived probs), and the cost of evasion across firms.
- ▶ **Differences over time:** Tight standard errors suggest that tax evasion heterogeneity is constant over time.
 - ▶ This might imply that the distributions of the firm characteristics, perceived probability of detection, and cost of evasion are constant over time.

Tax Evasion Over Time (Continued)

The 1983 fiscal reform changed incentives to evade taxes over time *differently* for firms in ST-liable and ST-exempt industries,

- ▶ Most ST-liable industries faced a Sales Tax (ST) rate *increase* from 6% to 10%.
- ▶ ST-exempt industries *remained exempt* before and after the reform.

Note that firms in *all* industries are subject to Corporate Income Tax (CIT).

- ▶ CIT also generates incentives to evade taxes by overreporting input costs.
 - ▶ Lower reported profits due to higher input costs reduce the tax base, and thus the tax liability.
- ▶ Therefore, CIT and ST generate tax evasion incentives for firms in ST-liable industries, while only CIT for firms in ST-exempt industries.

Unincorporated Firms by Industry Tax Liability

For unincorporated firms, the ST and CIT generated different incentives to evade taxes depending on whether they belong to an ST-exempt industry or not,

$$s_{ijt}^N = \gamma_j + e_{ijt}^{ST} + e_{ijt}^{CIT} - \varepsilon_{ijt} \quad j \in \text{ST-Liable} \quad (14)$$

$$s_{ijt}^N = \gamma_j + e_{ijt}^{CIT} - \varepsilon_{ijt} \quad j \in \text{ST-Exempt} \quad (15)$$

where e_{ijt}^{ST} and e_{ijt}^{CIT} stand for the overreporting due to ST and CIT, respectively.

Combining All Firms

- ▶ Let D_{ijt}^E be a dummy indicating whether the industry is ST-exempted; and
- ▶ D_{ijt}^L , a dummy indicating whether the industry is ST-liable
 - ▶ such that $D_{ijt}^E + D_{ijt}^L = 1$.

Unincorporated Firms in ST-Liable and ST-Exempt Industries

Combining Equation 15 with Equation 14,

$$\begin{aligned} D_{ijt}^E s_{ijt}^N + D_{ijt}^L s_{ijt}^N &= D_{ijt}^E (\gamma_j + e_{ijt}^{CIT} - \varepsilon_{ijt}) \\ &\quad + D_{ijt}^L (\gamma_j + e_{ijt}^{ST} + e_{ijt}^{CIT} - \varepsilon_{ijt}) \\ (D_{ijt}^E + D_{ijt}^L) s_{ijt}^N &= (D_{ijt}^E + D_{ijt}^L) \gamma_j + D_{ijt}^E e_{ijt}^{CIT} \\ &\quad + D_{ijt}^L (e_{ijt}^{ST} + e_{ijt}^{CIT}) - (D_{ijt}^E + D_{ijt}^L) \varepsilon_{ijt} \\ s_{ijt}^N &= \gamma_j + D_{ijt}^E e_{ijt}^{CIT} + D_{ijt}^L (e_{ijt}^{ST} + e_{ijt}^{CIT}) - \varepsilon_{ijt} \end{aligned} \tag{16}$$

Corporations and Unincorporated Firms

Likewise, combining Equation 7 with Equation 16,

$$\begin{aligned}(1 - D_{ijt}^N)s_{ijt} + D_{ijt}^N s_{ijt} = & (1 - D_{ijt}^N)(\gamma_j - \varepsilon_{ijt}) \\ & + D_{ijt}^N \left(\gamma_j + D_{ijt}^E e_{ijt}^{CIT} \right. \\ & \left. + D_{ijt}^L (e_{ijt}^{ST} + e_{ijt}^{CIT}) - \varepsilon_{ijt} \right) \quad (17) \\ s_{ijt} = & \gamma_j + D_{ijt}^N D_{ijt}^E e_{ijt}^{CIT} \\ & + D_{ijt}^N D_{ijt}^L (e_{ijt}^{ST} + e_{ijt}^{CIT}) - \varepsilon_{ijt}\end{aligned}$$

Taking Expectations

$$E[s_{ijt}|D^N = 0] = \gamma_j$$

$$E[s_{ijt}|D^N = 1, D^E = 1, D^L = 0] - E[s_{ijt}|D^N = 0] = \mu_E^{CIT}$$

$$E[s_{ijt}|D^N = 1, D^E = 0, D^L = 1] - E[s_{ijt}|D^N = 0] = \underbrace{\mu_L^{ST} + \mu_L^{CIT}}_{\mu_L^{ST+CIT}}$$

we can identify the average overreporting by firms in ST-exempt, μ_E^{CIT} , and in ST-liable industries, μ_L^{ST+CIT}

If we assume $\mu_E^{CIT} = \mu_L^{CIT}$, we can also identify the overreporting due to ST, μ_L^{ST} .

Changes Over Time

In the case of the fiscal policy changes, we might be interested in the changes in average tax evasion after the 1983 fiscal reform.

We can take the difference of the expectations of year t' after the reform and the reform year, $t = 1983$,

$$\begin{aligned} E[s_{ijt'} - s_{ijt=1983} | D^N = 1, D^E = 1, D^L = 0] &= \mu_{E,t'}^{CIT} - \mu_{E,t=1983}^{CIT} \\ &\equiv \Delta_{E,t'}^{CIT} \end{aligned}$$

$$\begin{aligned} E[s_{ijt'} - s_{ijt=1983} | D^N = 1, D^E = 0, D^L = 1] &= \mu_{L,t'}^{CIT+ST} - \mu_{L,t=1983}^{CIT+ST} \\ &\equiv \Delta_{L,t'}^{CIT+ST} \end{aligned}$$

In Regression Form

To estimate Equation 17, I use the following regression specification,

$$s_{ijt} = \gamma_j + \sum_{t=1981}^{1991} \mu_{E,t} D_{ijt}^N D_{ijt}^E + \sum_{t=1981}^{1991} \mu_{L,t} D_{ijt}^N D_{ijt}^L + \eta_{ijt} \quad (18)$$

- ▶ $\mu_{E,t}$ are the coefficients of input overreporting by firms in ST-exempt in year t , and $\mu_{L,t}$ of firms in ST-liaible industries
- ▶ γ_j , D_{ijt}^N , D_{ijt}^E , D_{ijt}^L , and η_{ijt} are defined as before.

Differences Over Time in Regression Form

$$\begin{aligned}s_{ijt} = & \gamma_j + \mu_0 D_{ijt}^N + \mu_1 D_{ijt}^N D_{ijt}^E + \mu_2 D_{ijt}^N D_{ijt}^L \\ & + \sum_{t=1981}^{1991} \mathbb{1}[t \neq 1983] \Delta_{E,t} D_{ijt}^N D_{ijt}^E \\ & + \sum_{t=1981}^{1991} \mathbb{1}[t \neq 1983] \Delta_{L,t} D_{ijt}^N D_{ijt}^L + \eta_{ijt}\end{aligned}\tag{19}$$

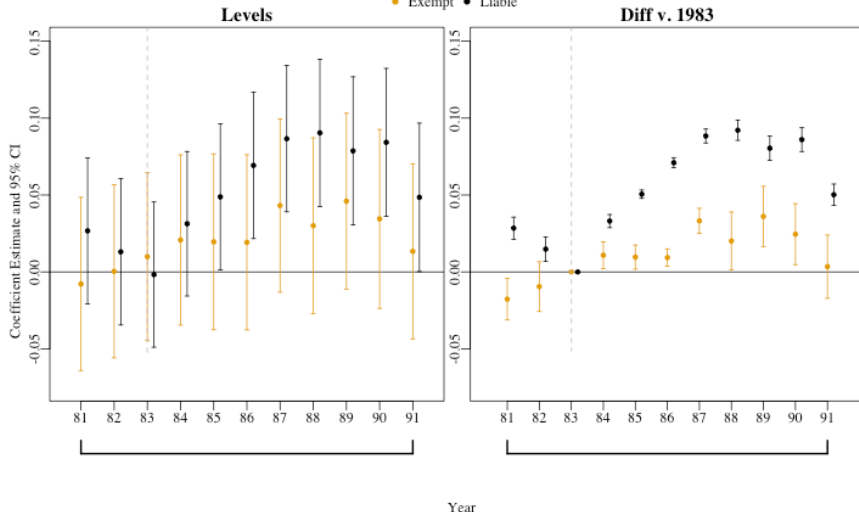
- ▶ $\Delta_{E,t}$ are the coefficients of the change in average tax evasion of firms in ST-exempt relative to 1983 and $\Delta_{L,t}$ of firms in ST-liable industries
- ▶ μ_0 , μ_1 , and μ_2 capture the average tax evasion of firms in 1983 of firms in both ST-liable and ST-exempt industries
- ▶ γ_j , D_{ijt}^N , D_{ijt}^E , D_{ijt}^L , and η_{ijt} are defined as before.

Firms Facing Larger Incentives Evade More

Tax Evasion by Unincorporated Firms

Input Overreporting in ST-Exempt and ST-Liable Industries

● Exempt ● Liable



Unincorporated Firms in Both Types of Industries

After the fiscal reform of 1983, we observe:

- ▶ Firms in ST-liable industries increased tax evasion gradually until 1987, where it stabilized until 1990
- ▶ Firms in ST-exempt industries show a moderate but significant increase in tax evasion after the fiscal reform of 1983, it then increased again between 1987 and 1990
- ▶ The increase is *larger* for ST-liable firms, the most affected by the fiscal reform of 1983

Splitting Unincorporated Firms by JO into LLCs and Proprietorships

Changes in Tax Evasion Over Time by Industry Type and Juridical Organization (JO)

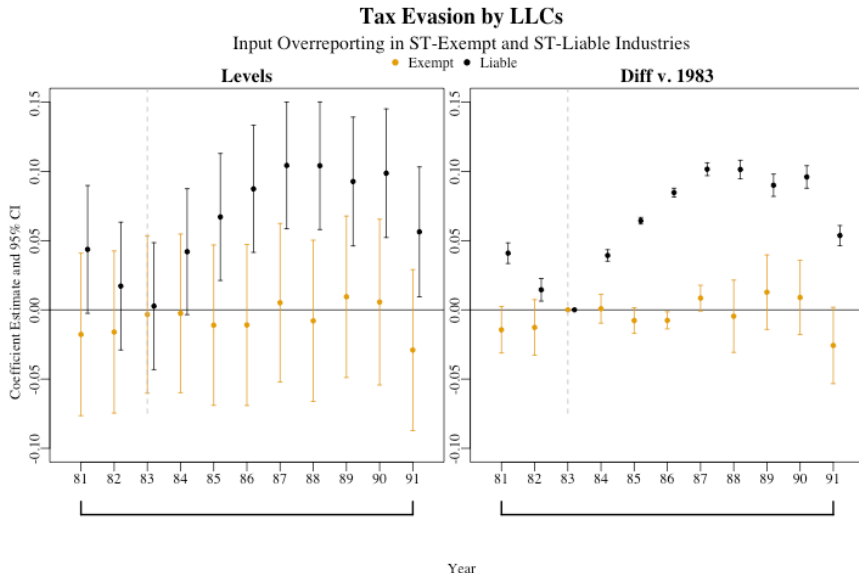
The tax evasion incentives generated by the 1983 fiscal policy reform also differed by JO,

- ▶ Differences *across* industries:
 - ▶ Most ST-liable industries faced a Sales Tax (ST) rate *increase* from 6% to 10%.
 - ▶ ST-exempt industries before the reform *remained exempt* after the reform.
- ▶ Differences *across* juridical organizations (JO) *within* industries:
 - ▶ Corporate Income Tax (CIT) increased 8% on average for most Proprietorships (PRTs) (through Individual Income Tax), while
 - ▶ CIT decreased 2 percentage points, from 20% to 18%, for Limited Liability Companies (LLCs)

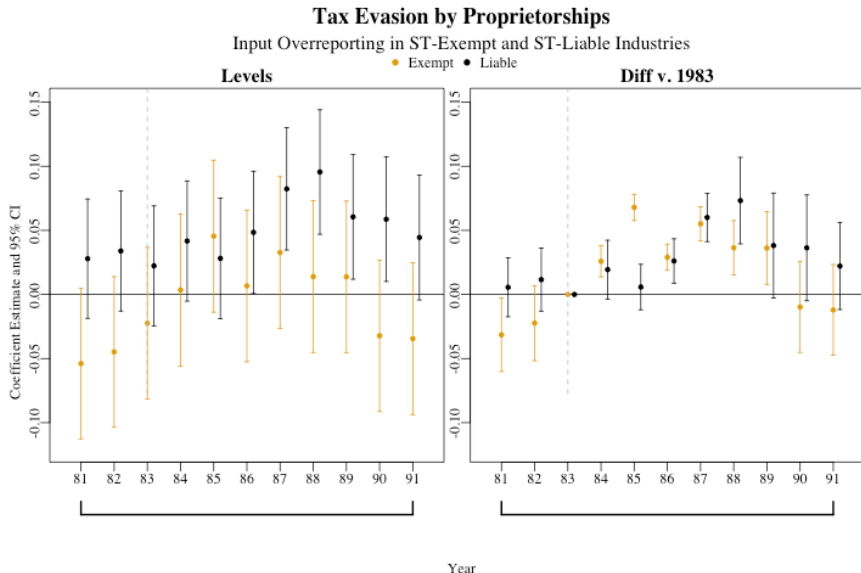
Expected Changes in Tax Evasion After the Fiscal Reform of 1983

- ▶ Since the **ST** rate *increased* for most ST-liable industries, we expect an *increase* in average *tax evasion* by these firms
- ▶ With respect to **CIT**, the expected change is *ambiguous*
 - ▶ Proprietorships faced an *increase* in their incentives to evade taxes, while
 - ▶ LLCs a small *decrease*.

LLCs with Larger Incentives Evade More



Proprietorships Response is More Heterogeneous



LLCs vs Proprietorships

- ▶ LLCs in ST-liable industries show a significant increase in tax evasion after the fiscal reform of 1983, consistent with the increase in their incentives to evade taxes due to the increase in the ST rate
- ▶ LLCs in ST-exempt industries show no significant change, suggesting that the small decrease in CIT for LLCs did not generate a significant decrease in tax evasion

LLCs vs Proprietorships (Cont.)

- ▶ Proprietorships in ST-exempt industries show a moderate but significant increase in tax evasion, consistent with the increase in their incentives to evade taxes due to the increase in CIT
- ▶ Proprietorships in ST-liable industries show no immediate change in tax evasion (1984-1985), but an increase between 1986 and 1989
- ▶ Overall, Proprietorships show a more heterogeneous response, as shown by the larger standard errors.
 - ▶ This makes sense, Proprietorships are likely to be more heterogeneous in characteristics (grandpa's workshop vs. young entrepreneurs, varying levels of process standardization, etc)

Takeaways

- ▶ Evidence suggests firms increase tax evasion by overreporting their inputs (materials) in response to increases in CIT and ST
- ▶ Overall, firms facing *larger increases* in ST and CIT tax rates evade more by overreporting a larger share of their input costs
- ▶ Firms facing a decrease in tax rates, such as LLCs in ST-exempt industries, show no significant decrease in tax evasion.
 - ▶ This finding suggests that firms might respond *asymmetrically* to changes in their incentives to evade taxes.

Tax Evasion and Firm Heterogeneity

Tax Evasion and Size

- ▶ Are there systematic differences in tax evasion across firms with different characteristics?
- ▶ Why? Firms might face different probabilities of detection, cost of evading taxes, and even different tax rates —thus, different incentives— depending on their characteristics.
- ▶ Size in particular:
 - ▶ Common intuition is that larger firms face higher probabilities of detection, and thus evade less.
 - ▶ However, recent evidence suggests that tax evasion increases along the size distribution (revenue) and then drops at the top of the distribution (Carrillo et al. 2022).

Empirical Strategy

Starting from FOC, as before, for corporations, the truth-reporters, we have,

$$s_{ijt}^C = \gamma_j - \varepsilon_{ijt} \quad (20)$$

Having recovered $\hat{\gamma}_j$ from the Corporations, we can use it to estimate the tax evasion of unincorporated firms, the evaders.

We can form,

$$\mathcal{V}_{ijt} \equiv s_{ijt} - \hat{\gamma}_j = e_{ijt} - \varepsilon_{ijt}$$

Empirical Strategy (Cont.)

- ▶ The objective is to find systematic differences in tax evasion between firms with different characteristics
- ▶ Empirically, we would like to estimate the expected tax evasion conditional on some size measure, $E[e_{ijt}|x_{ijt}]$
- ▶ The error term is theoretically independent of most firm characteristics that could influence tax evasion, $E[\varepsilon_{ijt}|x_{ijt}] = 0$, such as labor, capital, exports, imports, and even last period revenue
- ▶ Current revenue however is not independent of the error term, as it is part of the firm's output, and thus is correlated with the error term of the production function

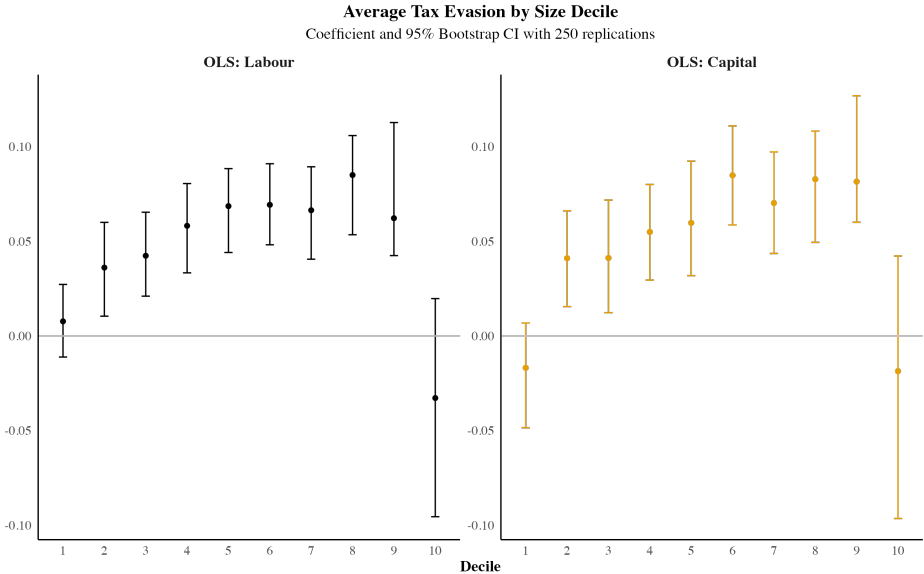
Regression Specification

First, I use the following regression specification to estimate the expected tax evasion conditional on size deciles,

$$\mathcal{V}_{ijt} = \sum_{h=1}^{10} \mathbb{1}[x_{ijt} \in \mathcal{X}_h] \mu_h + \eta_{ijt} \quad (21)$$

- ▶ x_{it} are the firm's decile of size measured as either labor, capital, current revenue, and previous year revenue ($t - 1$)
 - ▶ μ_h , the coefficients and capture the average tax evasion of firms in each decile of size
- ▶ γ_j , industry fixed effects, and η_{ijt} the error term, includes the component of tax evasion that is not captured by the size deciles, ξ_{ijt} and ε_{ijt}
- ▶ I use revenue _{$t-1$} as instrument for current revenue to correct for the endogeneity with ε_{ijt}

Tax Evasion Increases Along the Size Distribution and Then Drops at the Top



Tax Evasion Increases Along the Size Distribution and Then Declines Towards the Top

Average Tax Evasion by Size Decile

Coefficient and 95% Bootstrap CI with 250 replications

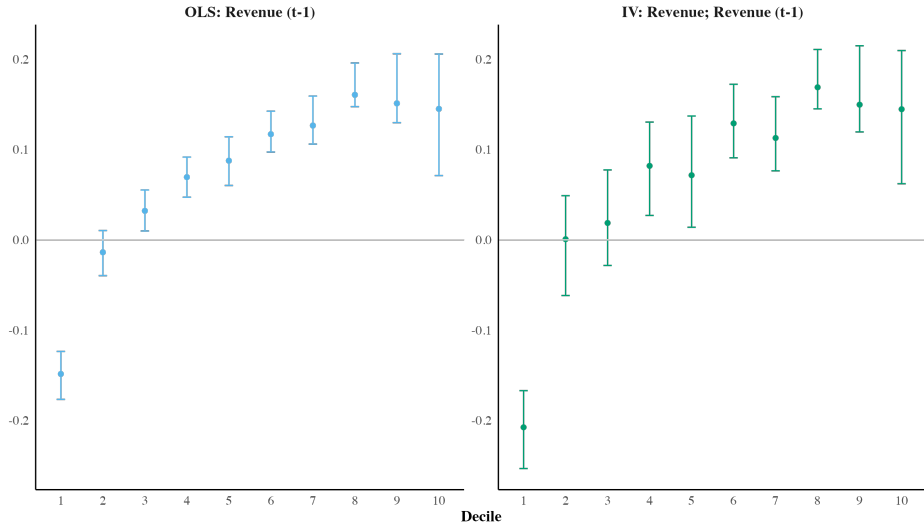


Table 1: Tax evasion estimates by size decile. 95% Bootstrap CI with 250 replications. Stars denote Bootstrap p-value significance at (***) 0.001, (**) 0.05, and (*) 0.1 level.

Decile	OLS: Labour	OLS: Capital	OLS: Revenue _{t-1}	IV: Revenue; Revenue _{t-1}
1	0.0077 [-0.0111, 0.0272]	-0.0168 [-0.0485, 0.0069]	-0.1485*** [-0.1766, -0.1235]	-0.2075*** [-0.2531, -0.1669]
2	0.0361*** [0.0105, 0.06]	0.0411*** [0.0155, 0.0661]	-0.0138 [-0.0396, 0.0103]	7e-04 [-0.0616, 0.0489]
3	0.0424*** [0.021, 0.0654]	0.0412*** [0.0123, 0.0718]	0.0321*** [0.0098, 0.0551]	0.0187 [-0.0284, 0.0774]
4	0.0582*** [0.0334, 0.0805]	0.055*** [0.0295, 0.0801]	0.0694*** [0.0471, 0.0915]	0.0819*** [0.027, 0.1303]
5	0.0686*** [0.0441, 0.0884]	0.0598*** [0.0319, 0.0924]	0.0876*** [0.0601, 0.1141]	0.0715** [0.0139, 0.1371]
6	0.0693*** [0.0482, 0.091]	0.0849*** [0.0587, 0.111]	0.1169*** [0.0972, 0.1424]	0.1289*** [0.0908, 0.1722]
7	0.0664*** [0.0406, 0.0894]	0.0703*** [0.0436, 0.0973]	0.1265*** [0.106, 0.1591]	0.1127*** [0.0764, 0.1585]
8	0.085*** [0.0535, 0.1059]	0.0828*** [0.0495, 0.1082]	0.1605*** [0.1474, 0.1957]	0.1688*** [0.145, 0.2107]
9	0.0622*** [0.0425, 0.1128]	0.0816*** [0.0601, 0.1269]	0.151*** [0.1297, 0.206]	0.1496*** [0.1195, 0.2148]
10	-0.0328 [-0.0955, 0.0197]	-0.0186 [-0.0965, 0.0423]	0.145*** [0.0711, 0.2057]	0.1446*** [0.0621, 0.2095]

Table 2: Median of Different Characteristics by Corporation and Top Three Deciles of Unincorporated Firms

Size	Characteristic	Corp	10th	9th	8th
Labour	Revenue (log)	13.1	12.9	11.8	11.1
	Capital (log)	11.7	11.7	10.5	9.9
	Labour (log)	5.6	6.1	5.2	4.7
	Materials (log)	12.6	12.4	11.3	10.7
	Imports (%)	0.2	0.1	0.0	0.0
	Exports (%)	0.0	0.0	0.0	0.0
Capital	Revenue (log)	13.1	13.2	12.0	11.4
	Capital (log)	11.7	12.3	11.1	10.4
	Labour (log)	5.6	5.9	4.8	4.5
	Materials (log)	12.6	12.7	11.6	11.0
	Imports (%)	0.2	0.1	0.0	0.0
	Exports (%)	0.0	0.0	0.0	0.0
Revenue	Revenue (log)	13.1	13.6	12.4	11.7
	Capital (log)	11.7	11.8	10.9	10.2
	Labour (log)	5.6	5.7	5.0	4.7
	Materials (log)	12.6	13.3	12.1	11.3
	Imports (%)	0.2	0.0	0.0	0.0
	Exports (%)	0.0	0.0	0.0	0.0

Tax Evasion and Size

- ▶ Similar pattern across size measures: Tax evasion increases along the size distribution and then drops/declines at the top of the distribution
- ▶ Surprisingly consistent with recent evidence in the literature using more detailed tax admin data (Figure 1).
- ▶ Large firms evade more than small firms, but the largest firms do not evade taxes.

But Why?

- ▶ Since the probability of detection cannot be, by logic, decreasing in size,
 - ▶ Most likely it is flat along the size distribution, and then increases rapidly at the top of the distribution
- ▶ It then follows that the cost of evasion should be decreasing along the size distribution, and then it might (or might not) increase at the top of the distribution.

Why the negative sign for some small firms?

Takeaways

Overall, these results suggest that the non-linear pattern observed in the previous section might be explained by:

1. a probability of detection that remains flat (or increases slowly) along most of the size distribution, but then increases sharply for the largest firms.
 - ▶ The detection probability increases sharply for the largest firms because they are more visible to the tax authority, and because
 - ▶ these firms are more likely to be denounced to the tax authority by an employee

Takeaways (Cont.)

2. a cost of evasion that decreases along the size distribution (but then it might increase rapidly for the largest firms). This is because
 - ▶ firms with small revenues face higher relative costs of evasion as they have fewer resources to allocate to tax evasion. As firms increase in revenue, they can allocate more resources to tax evasion, reducing the relative cost of evasion.
 - ▶ For the largest revenue firms however, the cost of evasion might increase abruptly due to the higher costs of illegal evasion relative to legal avoidance strategies accessible mostly for the largest firms only, and higher reputational costs if caught evading taxes.

The Geography of Tax Evasion

How Does a Firm's Location Affect its Tax Evasion?

Firms located in different regions might face different probabilities of detection and different tax rates, and thus different incentives to evade taxes.

► Tax rates:

- Departments and municipalities in Colombia can set their own tax rates similar to sales tax, for example the Industry and Commerce Tax
- Special tax regimes for firms located in certain regions, such as the Free Trade Zones (Zonas Francas) and the Special Economic Zones (Zonas Económicas Especiales)
- Special ownership model in Atlantic coast
- Some variation in tax rates is observed in the data

► Probability of detection: Two hypothesis

1. Firms located near the tax authority's offices might have access to the tax authority (lobbying, bribery, etc.) and thus face a lower probability of detection
2. Firms located farther away from the tax authority's offices might face a lower probability of detection, because the auditing cost for the tax authority increases with distance, and thus they are less likely to be audited

Data

In the dataset from Colombia, I have two geographic variables:

- ▶ `section_country_code`: Section of the country (eg. Antioquia, Atlantico, Bogota D.E., etc.)
- ▶ `metro_area_code`: Metropolitan Area (eg. Bogota, Cali, Medellin, Barranquilla).

Metro Area Summary Statistics

Metro Area	N	Market Share	Avg. Sales Tax Rate	Med. Sales Tax Rate	Avg. Tax Evasion	Med. Tax Evasion
Bogotá D.C., Soacha	14345	35.3	7.4	8.9	4.2	10.3
Medellín, Valle de Aburrá	8147	19.9	7.6	9.2	4.6	10.7
Rest of Colombia	5895	14.9	4.6	2.0	0.7	10.2
Cali, Yumbo	4538	11.4	7.2	8.4	4.2	10.1
Barranquilla, Soledad	2896	7.3	5.6	5.9	2.5	13.3
Bucaramanga, Giron, Floridablanca	2311	5.3	5.7	6.0	5.9	13.8
Pereira, Santa Rosa de Cabal, Dosquebradas	1128	2.7	7.6	9.1	11.1	21.1
Manizales, Villamaria	639	1.6	7.0	8.1	3.2	7.2
Cartagena	606	1.6	4.6	2.7	11.6	20.0

Section of the Country Summary Statistics

Depto	N	Market Share	Avg. Sales Tax Rate	Med. Sales Tax Rate	Avg. Tax Evasion	Med. Tax Evasion
Bogotá D.C.	13954	34.3	7.3	8.9	4.3	10.3
Antioquia	8467	20.7	7.5	9.1	3.4	9.7
Valle del Cauca	5780	14.6	6.7	7.0	4.1	10.3
Atlántico	2922	7.3	5.6	5.8	2.6	13.2
Santander	2585	6.0	5.6	6.0	5.5	12.8
Cundinamarca	1282	3.5	5.8	6.0	1.9	12.8
Risaralda	1161	2.8	7.4	8.9	11.0	21.0
Tolima	763	1.9	4.0	0.0	10.1	21.2
Caldas	710	1.8	6.7	6.8	-0.8	6.6
Bolívar	635	1.7	4.4	2.1	9.8	19.7
Norte de Santander	617	1.4	5.2	6.0	-7.6	2.9
Boyacá	290	0.7	5.8	6.0	1.8	8.8
Quindío	288	0.7	5.2	6.0	6.9	18.1
Nariño	255	0.6	3.4	0.0	14.7	25.1
Huila	182	0.5	4.4	0.0	10.1	19.2
Cauca	155	0.4	3.7	0.0	-0.4	-1.8
Meta	119	0.3	2.8	0.0	-10.5	12.4
Córdoba	126	0.3	3.7	0.3	21.1	25.1
Cesar	72	0.2	1.8	0.0	-11.8	16.9
Magdalena	73	0.2	5.0	1.3	-33.7	-13.2
Sucre	54	0.1	4.5	4.4	-5.7	-11.4
Putumayo	9	0.0	8.4	6.0	22.8	7.1
Caquetá	5	0.0	5.9	6.0	50.7	70.1
Chocó	1	0.0	3.1	3.1	-36.0	-36.0

Prelim Results (Metro Area)

Dependent Variable:	$\mathcal{V}$
Model:	(1)
<i>Variables</i>	
Bogotá D.C., Soacha	0.0485*** (0.0132)
Cali, Yumbo	0.0706*** (0.0174)
Medellín, Valle de Aburrá	0.0440** (0.0141)
Manizales, Villamaria	0.0359 (0.0512)
Barranquilla, Soledad	0.0309 (0.0266)
Bucaramanga, Giron, Floridablanca	0.0654** (0.0226)
Pereira, Santa Rosa de Cabal, Dosquebradas	0.1391*** (0.0355)
Cartagena	0.1402* (0.0758)
Rest of Colombia	0.0244 (0.0197)
<i>Fit statistics</i>	
Observations	32,373
R ²	0.00280
Adjusted R ²	0.00255

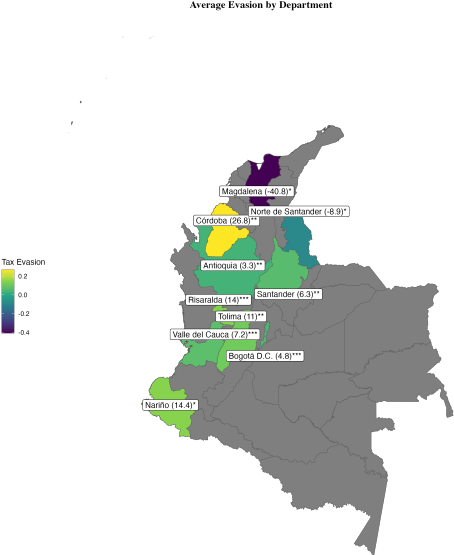
Clustered (plant & year) standard-errors in parentheses
*Signif. Codes: ***, 0.01, **, 0.05, *, 0.1*

Prelim Results (Departamento)

Dependent Variable: Model:	ψ (1)
<i>Variables</i>	
Antioquia	0.0331** (0.0143)
Atlántico	0.0323 (0.0267)
Bogotá D.C.	0.0484*** (0.0133)
Bolívar	0.1116 (0.0743)
Boyacá	0.0498 (0.0822)
Caldas	-0.0275 (0.0574)
Cauca	0.1291 (0.0779)
Cesar	-0.3661 (0.3888)
Córdoba	0.2684** (0.0882)
Cundinamarca	0.0371 (0.0449)
Huila	0.1464 (0.0816)
Magdalena	-0.4084* (0.2023)
Meta	-0.1142 (0.1742)
Nariño	0.1441* (0.0655)
Norte de Santander	-0.0894* (0.0427)
Quindío	0.0801 (0.0608)
Risaralda	0.1396*** (0.0349)
Santander	0.0629** (0.0227)
Sucre	0.0642 (0.1590)
Tolima	0.1092** (0.0414)
Valle del Cauca	0.0718*** (0.0152)
<i>Fit statistics</i>	
Observations	32,373
R ²	0.00895
Adjusted R ²	0.00834

Clustered (plant & year) standard errors in parentheses
 Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

On the Map



Next Steps

- ▶ Are there any patterns? Do they make sense?
- ▶ Can any of these regions be linked to Free Trade Zones or Special Economic Zones? Since when are these in place?

Couterfactual Question

What Would Be The Optimal Tax Rate?

The government faces a trade-off:

- ▶ as the government increases the tax rate, the tax revenue increases
- ▶ as the tax rate increases, the marginal benefit of evading taxes increase leading to higher levels of tax evasion which reduces tax revenue

What is the optimal rate that maximizes expected tax revenue for the government?

What is the tax rate that lead to median (media) tax evasion of zero?

The Government's Problem

The government wants to choose the sales tax rate τ that maximizes the expected tax revenue from the sum of all firm i

$$\max_{\tau} \sum_i q(e, a) \tau [P_t Y_{ijt} - \rho_t M_{ijt}] + (1 - q(e, a)) \tau [P_t Y_{ijt} - \rho_t (M_{ijt} + e_{ijt})]$$

Simplifying,

$$\max_{\tau} \sum_i \tau [P_t Y_{ijt} - \rho_t M_{ijt} - (1 - q(e, a)) \rho_t e_{ijt}]$$

where $q(e, a)$ is the probability of detection, which is a function of the evasion level, e , and the government effort a .

for firm i ,

$$\begin{aligned} & [P_t Y_{ijt} - \rho_t M_{ijt} - (1 - q(e, a)) \rho_t e_{ijt}] \\ & + \tau \left[-(1 - q(e, a)) \rho_t \frac{\partial e_{ijt}}{\partial \tau} + \rho_t e_{ijt} \frac{\partial q(e, a)}{\partial e} \frac{\partial e_{ijt}}{\partial \tau} \right] = 0 \end{aligned}$$

simplifying,

$$\begin{aligned} P_t Y_{ijt} - \rho_t M_{ijt} &= (1 - q(e, a)) \rho_t e_{ijt} \\ &+ \tau \rho_t \frac{\partial e_{ijt}}{\partial \tau} \left[(1 - q(e, a)) - \frac{\partial q(e, a)}{\partial e} e_{ijt} \right] \end{aligned}$$

Numerical Strategy

1. Define functional forms for the probability of detection, $q(e, a)$, and the cost of evasion, $\kappa(e)$, that are consistent with the observed patterns in the data and with theory
2. Estimate the parameters of these functions using moments from the FOC of the firm's optimization problem (Equation 4) and the deconvolved distribution of tax evasion

3. Given the estimated parameters and a proposed tax rate, find the simulated distribution of tax evasion $f_{\tilde{e}}$ such that
 - i. the FOC wrt e of the firm's optimization problem (Equation 4) holds on average with the new τ and
 - ii. the moments derived from $E[\varepsilon|\tilde{e}] = 0$ also hold (e.g $E[\varepsilon\tilde{e}] = 0$, $E[\varepsilon\tilde{e}^2] = 0$, etc.)
4. Compute the expected tax revenue for the proposed tax rate using the simulated distribution of tax evasion
 - i. Note that since I have $f_e(e)$, I also have the distribution of the true inputs since $M = M^* - e$, and thus I can compute the expected tax revenue for a simulated tax evasion $\tilde{e}$ corresponding to the proposed $\tilde{\tau}$

$$E[R(\tilde{\tau})] = \sum_i \left[\tilde{\tau} \int [P_t Y_{ijt} - \rho_t(M_{ijt}^* - e)] f_e(e) de - \tilde{\tau} \int (1 - q(\tilde{e}, a)) \rho_t \tilde{e} f_{\tilde{e}}(\tilde{e}) d\tilde{e} \right]$$

5. Find the tax rate that maximizes expected tax revenue

Functional Forms

Setting the effect of size aside for the moment, I know the following properties of the probability of detection, $q(e, a)$, and the cost of evasion, $\kappa(e)$, functions:

- ▶ $q(e, a) \in [0, 1]$, $q(0, a) = 0$, $q(\infty, a) = 1$, and $q(e, a)$ is increasing in e .
Since I do not have variance in govt effort, a will be constant $q(e)$
 - ▶ $q(e) = 1 - \exp\{-\lambda e\}$
- ▶ $\kappa(e) \in [0, \infty)$, $\kappa(0) = 0$, $\kappa'(0) \geq 0$, and $\kappa(e)$ is increasing in e
 - ▶ $\kappa(e) = \delta e$
 - ▶ Maybe the cost varies between Corporations and Unincorporated firms, $\kappa(e, O) = e(\delta_0 + \mathbb{1}\{O = \text{Corp}\}\delta_1)$, where O is the firm's juridical organization. In this way, we can allow for Corporations to select the corner solution of no evasion, while Unincorporated firms select an interior solution with positive evasion. Corp: $\tau\rho_t \leq \delta_0 + \delta_1$; Unicorp:
 $\tau\rho_t(1 - (1 - \exp\{-\lambda e\}) - \lambda e \exp\{-\lambda e\}) = \delta_0$

Simplifying FOC of Unicorps, $\tau\rho_t \exp\{-\lambda e\}(1 + \lambda e) = \delta_0$

Estimation and Data

- From data, I have sales tax paid on purchases $\tau\rho_t M_{ijt}^*$ and reported intermediates M_{ijt}^* at the firm level.

Using these variables from the data and taking logs,

$$\ln(\tau\rho_t M_{ijt}^*) - \lambda e_{ijt} + \ln(1 + \lambda e_{ijt}) = \ln(M_{ijt}^* \delta_0)$$

$$\ln(\tau\rho_t M_{ijt}^*) - \ln(M_{ijt}^*) = \ln(\delta_0) + \lambda e_{ijt} - \ln(1 + \lambda e_{ijt})$$

$$\ln\left(\frac{\tau\rho_t M_{ijt}^*}{M_{ijt}^*}\right) = \ln(\delta_0) + \lambda e_{ijt} - \ln(1 + \lambda e_{ijt})$$

Am I missing an error term here?

- ▶ Measurement error when reporting data on sales tax paid on purchases
- ▶ Idiosyncratic shocks to the cost of evasion across firms and over time.
- ▶ Since I know $f_e(e)$, I could deconvolve the distribution of this error term, this error term should be:
 1. Independent of true materials M_{ijt} and
 2. Independent of the tax evasion level e_{ijt} ,
 3. If not independent of e , Can I use the fact that $f_{e,\xi}(e, \xi) = f_{e|\xi}(e|\xi)f_\xi(\xi) = f_{\xi|e}(\xi|e)f_e(e)$, so I can get away deconvolving the distribution of the error term conditional on e instead of the unconditional distribution of the error term?

Implicit Assumptions

- ▶ The functional forms for $q(e)$ and $\kappa(e)$ are correctly specified and do not change over time
- ▶ No entry, no exit
- ▶ Corporations never evade (?)
- ▶ Partial equilibrium, Inelastic demand, (No GE effects)
 - ▶ Tax increases do not change optimal input choices and output. FOC of materials does not depend on tax rate
 - ▶ Firms are already assumed to be price takers

Appendix

Appendix

B) Value of Reported Ghost Purchases Over Firm Total Purchases, by Firm Size

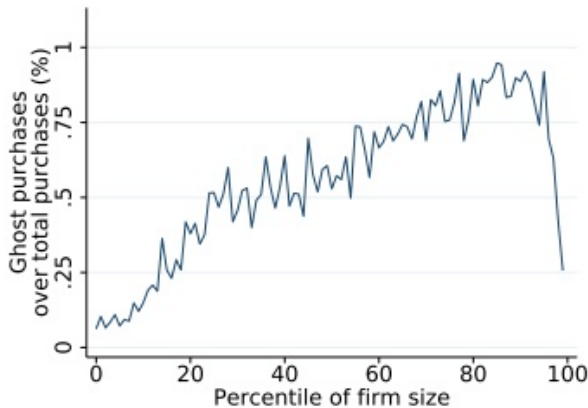


Figure 1: Figure 1B from Carrillo et al. (2022) ↵

Robust Check for Size Heterogeneity

$$\mathcal{V}_{ijt} = \sum_{z \in Z} \sum_{h \in 1,2} \mu_{z,h} x_{z,ijt}^h + \eta_{ijt} \quad (22)$$

Table 3: Coefficient estimates . 95% Bootstrap CI with 250 replications. Stars denote Bootstrap p-value significance at (***) 0.001, (**) 0.05, and (*) 0.1 level.

	IV: Lagged Revenue	OLS: Lagged Revenue
Revenue	1.8012*** [1.5744, 2.0744]	
Revenue ²	-0.7594*** [-1.0265, -0.5149]	
Revenue _{t-1}		1.456*** [1.274, 1.6735]
Revenue _{t-1} ²		-0.5586*** [-0.784, -0.339]
Labour	-0.7149*** [-0.9244, -0.5631]	-0.4281*** [-0.6214, -0.2994]
Labour ²	0.1051 [-0.0561, 0.3035]	-0.09 [-0.2306, 0.1051]
Capital	-0.1124 [-0.277, 0.085]	-0.022 [-0.1776, 0.1737]
Capital ²	-0.2332** [-0.4548, -0.0406]	-0.2997*** [-0.5255, -0.1106]
Exports	-0.1127 [-0.7196, 0.3843]	-0.2759 [-0.8917, 0.192]
Exports ²	0.066 [-0.483, 0.7506]	0.2503 [-0.2569, 0.9386]
Imports	-1.1048*** [-1.4765, -0.7346]	-1.1253*** [-1.5047, -0.7651]
Imports ²	1.3084*** [0.8792, 1.7581]	1.3469*** [0.9235, 1.8055]
Const.	-0.1309*** [-0.1662, -0.0985]	-0.1158*** [-0.1492, -0.0826]

Table 4: Coefficient estimates. Percentile of size of different variables. 95% Bootstrap CI with 250 replications. Stars denote Bootstrap p-value significance at (***) 0.001, (**) 0.05, and (*) 0.1 level.

	IV: Lagged Revenue	OLS: Lagged Revenue
Revenue	1.1491*** [1.0863, 1.2497]	
Revenue _{t-1}		0.9948*** [0.9362, 1.0738]
Labour	-0.6258*** [-0.7001, -0.5599]	-0.5044*** [-0.5691, -0.4402]
Capital	-0.3082*** [-0.3679, -0.2278]	-0.2737*** [-0.3309, -0.1943]
Exports	-0.1004 [-0.5756, 0.283]	-0.2292 [-0.7139, 0.1385]
Imports	-0.7292*** [-0.9883, -0.5182]	-0.7386*** [-1.0034, -0.5334]

Why the Negative Sign for Some Small Firms?

- ▶ Smaller firms are more likely to be Proprietorships and less formal
- ▶ Proprietorships report their industrial activity as part of their individual income taxes. They need to keep track of their personal expenses and their industrial activity expenses
- ▶ Because they are more likely to be less formal than larger firms, they may lack a rigorous expense report system
- ▶ Thus, they might be consistently underreporting their expenses

References

Carrillo, Paul, Dave Donaldson, Dina Pomeranz, and Monica Singhal. 2022.
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<http://www.nber.org/papers/w30242>.